

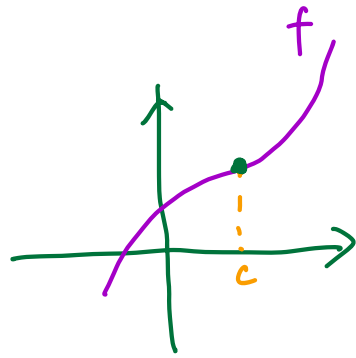
Lec 5-1 [Mon. Sep 29].

- Second derivative test (4.4)
- Concavity (4.4)
- Graph sketching (4.4)

Q: How do we find inflection points?

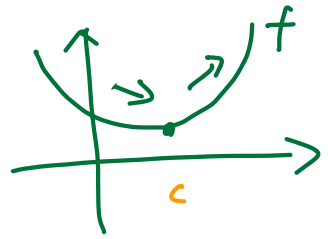
Thm. If c is an inflection point of f , then

either $f''(c)$ does not exist or $f''(c) = 0$.



Pf: [use the idea of Fermat's local extrema theorem] [Optional]
(Thm 4.1.2)

Step 1. write down the definition of an inflection point.



• To prove: if $f''(c)$ exists, then $f''(c) = 0$.

• By definition, if c is an inflection point of f , then

$\exists \delta > 0$, s.t. f changes the concavity from $(c-\delta, c)$ to $(c, c+\delta)$.

• Suppose f is concave down on $(c-\delta, c)$ and concave up on $(c, c+\delta)$

Step 2: compute the left and right derivatives.

we have f' is decreasing on $(c-\delta, c) \Rightarrow f'(x) \geq f'(c), \forall x \in (c-\delta, c)$.

Hence, we have
$$\frac{f'(x) - f'(c)}{x - c} \leq 0, \forall x \in (c-\delta, c). \quad (*)$$

We have f' is increasing on $(c, c+\delta) \Rightarrow f'(x) \geq f'(c), \forall x \in (c, c+\delta)$.

Hence, we have
$$\frac{f'(x) - f'(c)}{x - c} \geq 0, \forall x \in (c, c+\delta). \quad (**)$$

Step 3. Both ≥ 0 and ≤ 0 .

As $f''(c)$ exists finitely,

$$f''(c) = \lim_{x \rightarrow c^+} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \rightarrow c^-} \frac{f'(x) - f'(c)}{x - c}$$

$$\text{And } (*) \Rightarrow \lim_{x \rightarrow c^-} \frac{f'(x) - f'(c)}{x - c} \leq 0.$$

$$\text{And } (**) \Rightarrow \lim_{x \rightarrow c^+} \frac{f'(x) - f'(c)}{x - c} \geq 0.$$

$$\therefore f''(c) = 0.$$

□.

e.g.2. Determine all points of inflection for the curves:

(a) $y = x^{1/3}$, $D = \mathbb{R}$

(b) $y = x^4$, $D = \mathbb{R}$

(c) $y = x^{4/3} - 4x^{1/3}$, $D = \mathbb{R}$.

[Idea: $f''(c) = 0$ or $f''(c)$ does not exist!]

Sol: (a)

Step 1. Take the first derivative!

$$y' = \frac{1}{3} x^{-\frac{2}{3}}, \quad x \neq 0.$$

$$\therefore \lim_{x \rightarrow 0^+} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^{\frac{1}{3}}}{x} = \lim_{x \rightarrow 0^+} \frac{1}{x^{2/3}} = \infty.$$

$$\lim_{x \rightarrow 0^-} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{x^{\frac{1}{3}}}{x} = \lim_{x \rightarrow 0^-} \frac{1}{x^{2/3}} = \infty.$$

$\therefore y'(0)$ does not exist finitely.

$$\therefore y'(x) = \begin{cases} \frac{1}{3} x^{-\frac{2}{3}}, & x \neq 0 \\ \text{not defined}, & x = 0 \end{cases}$$

Step 2. Take the second derivative!

$$\therefore y''(x) = \begin{cases} -\frac{2}{9} x^{-\frac{5}{3}}, & x \neq 0; \\ \text{not defined}, & x = 0. \end{cases}$$

[as $y'(0)$ is not defined,

$\lim_{h \rightarrow 0} \frac{y'(x) - y'(0)}{x - 0}$ does not exist either.]

Step 3. solve $y''(c) = 0$!

$$\therefore y''(x) \neq 0, \quad \forall x \neq 0.$$

$\therefore$ There is no solution for $y''(c) = 0$.

$\therefore$ The only possible inflection point is 0.

Step 4 . check the definition of inflection point:

$$\left(-\frac{2}{9}\right) \frac{1}{(-0.1)^{5/3}}$$

e.g. $\delta = 0.1$

① . on the left side, $y''(x) = -\frac{2}{9} x^{-5/3} > 0$, $y' \uparrow$ on $(0-\delta, 0)$

· on the right side, $y''(x) = -\frac{2}{9} x^{-5/3} < 0$, $y' \downarrow$ on $(0, 0+\delta)$.

$\therefore f$ changes concavity at 0!

② f has a

$$\therefore \lim_{h \rightarrow 0^+} \frac{y(h) - y(0)}{h - 0} = \infty$$

$$\lim_{h \rightarrow 0^-} \frac{y(h) - y(0)}{h - 0} = \infty$$

$$\therefore \lim_{h \rightarrow 0} \frac{y(h) - y(0)}{h - 0} = \infty$$

$\therefore 0$ is the inflection point of f .

Q.E.D.

② $y = x^4, D = \mathbb{R}.$

Sol: Step 1-2. first derivative and second derivative

$$y'(x) = 4x^3, x \in \mathbb{R}.$$

$$y''(x) = 12x^2, x \in \mathbb{R}.$$

Step 3. Solve $y''(c) = 0.$

$$y''(c) = 12c^2 = 0 \Rightarrow c = 0.$$

Step 4. Check the definition of inflection point.

① y has a tangent line at $c=0$ ($\because y'(c)$ exists)

Let $\delta = 0.1.$

② $\wedge$ On the left side, $y'' > 0$ on $(-\delta, 0)$

On the right side, $y'' > 0$ on $(0, \delta).$

$\therefore y$ does not change the concavity at 0.

$\therefore y$ has no inflection point. □

Q: Is $y(x) = x^4$

concave up on $\mathbb{R}$?



A: Yes.

$\left[\begin{array}{l} \cdot y'(x) = 4x^3 \text{ is increasing} \Rightarrow \text{conc' up!} \\ \cdot y''(x) = 12x^2 \text{ may not always } > 0 \text{ (equal to 0 at 0)} \end{array} \right]$

③ $y = x^{4/3} - 4x^{1/3}$, $D = \mathbb{R}$.

Sol: Step 1. First derivative

• $y'(x) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3}$, $x \neq 0$.

• $y'(0)$ does not exist.

Step 2. Second derivative

• $y''(x) = \frac{4}{9}x^{-2/3} + \frac{8}{9}x^{-5/3}$
 $= \frac{4}{9}x^{-5/3}(x+2)$, $x \neq 0$.

• $y''(0)$ does not exist.

Step 3. Solve $y''(c) = 0$.

$$y''(c) = \frac{4}{9}c^{-5/3}(c+2) = 0 \Rightarrow c = -2$$

$\therefore -2$ and 0 are all possible inflection points.

Step 4. Check the definition of inflection points.

A) for -2 ,

let $\delta = 0.1$,

① on the left side, $y''(x) = \frac{4}{9}x^{-5/3}(x+2) > 0$ on $(-2-\delta, -2)$;
 $\wedge$ $\underbrace{x^{-5/3}}_{<0} \underbrace{(x+2)}_{<0}$

on the right side, $y''(x) = \frac{4}{9}x^{-5/3}(x+2) < 0$ on $(-2, -2+\delta)$.
 $\underbrace{x^{-5/3}}_{<0} \underbrace{(x+2)}_{>0}$

$\therefore y$ changes the concavity at -2 .

② y is differentiable at -2 ($\because y'(-2)$ exists finitely)

$\therefore -2$ is an inflection point.

B) for 0 ,

① on the left side, let $\delta = \epsilon$. $y''(x) = \frac{4}{9}x^{-\frac{5}{3}}(x+2) < 0$ on $(-\delta, 0)$;

on the right side, $y''(x) = \frac{4}{9}x^{-\frac{5}{3}}(x+2) > 0$ on $(0, \delta)$.

$\therefore y$ changes the concavity at 0 .

② y has a vertical tangent line at 0 .

$$\left[\because \lim_{x \rightarrow 0^+} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0^+} \frac{x^{\frac{4}{3}} - 4x^{\frac{1}{3}}}{x} = \lim_{x \rightarrow 0^+} \left(x^{\frac{1}{3}} - 4 \cdot \frac{1}{x^{\frac{2}{3}}} \right) = -\infty \right.$$

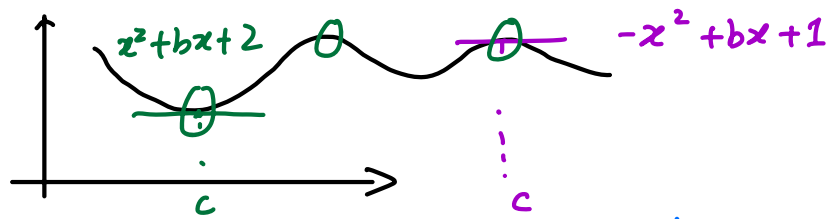
$$\left. \lim_{x \rightarrow 0^-} \frac{y(x) - y(0)}{x - 0} = \lim_{x \rightarrow 0^-} \frac{x^{\frac{4}{3}} - 4x^{\frac{1}{3}}}{x} = \lim_{x \rightarrow 0^-} \left(x^{\frac{1}{3}} - 4 \cdot \frac{1}{x^{\frac{2}{3}}} \right) = -\infty \right.$$

$\therefore 0$ is also an inflection point.

$\therefore -2, 0$ are inflection points.

□.

Second derivative test.



Thm 4.4.5. Suppose f'' exists for any point in an open interval that contains $x=c$.

① If $f'(c)=0$ and $f''(c)>0$, then f has a local $\min$ at $x=c$.

② If $f'(c)=0$ and $f''(c)<0$, then f has a local $\max$ at $x=c$.

③ If $f'(c)=0$ and $f''(c)=0$, then the test fails. The function f may have a local $\max$, a local $\min$ or neither.

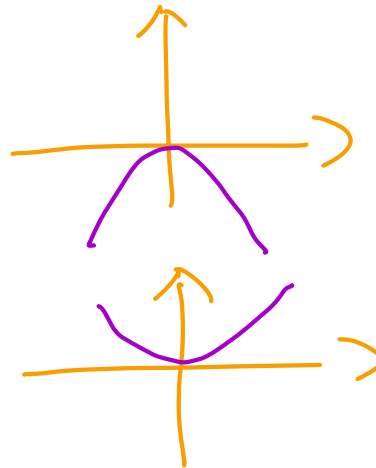
Memory :

① $f(x) = -x^2$ at 0.

$$f'(c)=0 \Rightarrow c=0$$

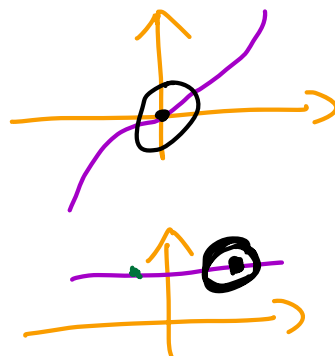
$$f''(c)=-2<0$$

② $f(x) = x^2$ at 0



③ $f(x) = x^3$

$$f(x) \equiv 1.$$



Pf of Thm 4.4.5 [Optional]

a) $f''(c) > 0 \Rightarrow f'$ is increasing locally

b) $f'(c) = 0 \Rightarrow f' < 0$ on left, $f' > 0$ on right
 $\Rightarrow f \searrow$ on left, $f \nearrow$ on right
 $\Rightarrow f$ locally min at c .

②. Step 1. estimate $f'(x)$ locally at c .

Suppose $f'(c) = 0$ and $f''(c) > 0$.

Let $L = f''(c) > 0$.

$$\therefore L = f''(c) = \lim_{x \rightarrow c} \frac{f'(x) - f'(c)}{x - c} = \lim_{x \rightarrow c} \frac{f'(x) - 0}{x - c}.$$

$\therefore$ For $\varepsilon = \frac{L}{2}$, $\exists \delta > 0$, s.t.

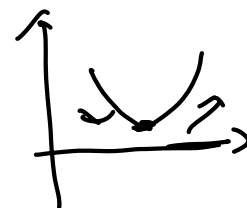
$$\left| \frac{f'(x) - 0}{x - c} - L \right| < \varepsilon, \quad \forall x \in (c - \delta, c + \delta) \setminus \{c\}.$$

$$\Rightarrow -\varepsilon < \frac{f'(x)}{x - c} - L < \varepsilon, \quad \forall x \in (c - \delta, c + \delta) \setminus \{c\}.$$

$$\Rightarrow 0 < \frac{L}{2} = L - \varepsilon < \frac{f'(x)}{x - c} < L + \varepsilon, \quad \forall x \in (c - \delta, c + \delta) \setminus \{c\}.$$

$$\Rightarrow f'(x) < 0, \quad \forall x \in (c - \delta, c)$$

and $f'(x) > 0, \quad \forall x \in (c, c + \delta)$



Step 2. use monotonicity

$\therefore f$ is increasing on $(c, c + \delta)$ and decreasing on $(c - \delta, c)$

$\therefore f$ has a local minimum at c .

The proof of ② is similar.

The proof of ③ is given by providing the above examples. $\square$.

e.g. 3. $f(x) = x^4 - 4x^3 + 10$. Find all local extrema using second derivative test.

Sol: Step 1. Solve all critical points and endpoints.

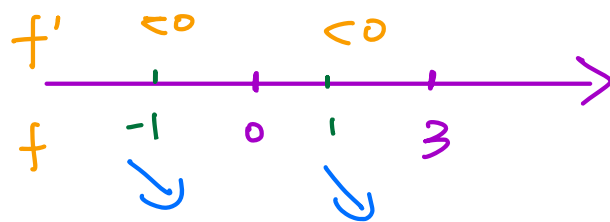
$$\cdot f'(x) = 4x^3 - 12x^2 = 4x^2(x-3)$$

$$\cdot f'(c) = 0 \Rightarrow c_1 = 0 \text{ or } c_2 = 3.$$

Besides, f is differentiable at any $x \in \mathbb{R}$
and there is no endpoint.

Step 2. Second derivative test

$$f''(x) = 12x^2 - 24x = 12x(x-2).$$



$$\therefore f''(3) = 12 \times 3 \times 1 > 0$$

$\therefore f$ has a local min at 3 and the value is $f(3) = -17$.

$$\therefore f''(0) = 12 \cdot 0 \cdot (-2) = 0.$$

$\therefore$ The second derivative test fails.

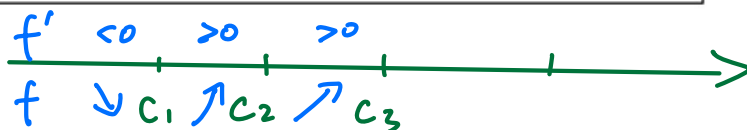
But we use first derivative test and know $f'(-1) < 0$, $f'(1) < 0$, which means f has no local extrema at 0. $\square$.

Summary for Local Extrema

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across c from left to right,

1. if f' changes from negative to positive at c , then f has a local minimum at c ;
2. if f' changes from positive to negative at c , then f has a local maximum at c ;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c .



4.4.5

THEOREM 4.4.5 — Second Derivative Test for Local Extrema

on an open interval that contains $x = c$.

Suppose f'' ~~is continuous~~ ^{exists}

1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.

Graph Sketching

Key components in sketching the graph $y = f(x)$.

- ① domain D and symmetry (even or odd function)
- ② asymptotes : horizontal, vertical, oblique and not-defined points $\rightarrow \infty$
 $\rightarrow -\infty$
 $\frac{1}{x}$
- ★ ③ monotonicity : critical points, intervals,
- ★ ④ concavity : critical points, intervals,
- ⑤ x- and y-intersects : points of intersection of this function with x- and y-axis.
- ⑥ f-values of key points.

e.g. sketch $y = f(x) = \frac{1}{x} - x^2$ $\frac{1}{-0.1} - (-0.1)^2 = -\infty$

① domain : $D = (-\infty, 0) \cup (0, \infty)$. Neither even or odd.

② asymptotes: 0 : $\lim_{x \rightarrow 0^+} f(x) = \infty$, $\lim_{x \rightarrow 0^-} f(x) = -\infty$. ∞ : $\lim_{x \rightarrow \infty} f(x) = -\infty$. $-\infty$: $\lim_{x \rightarrow -\infty} f(x) = -\infty$.

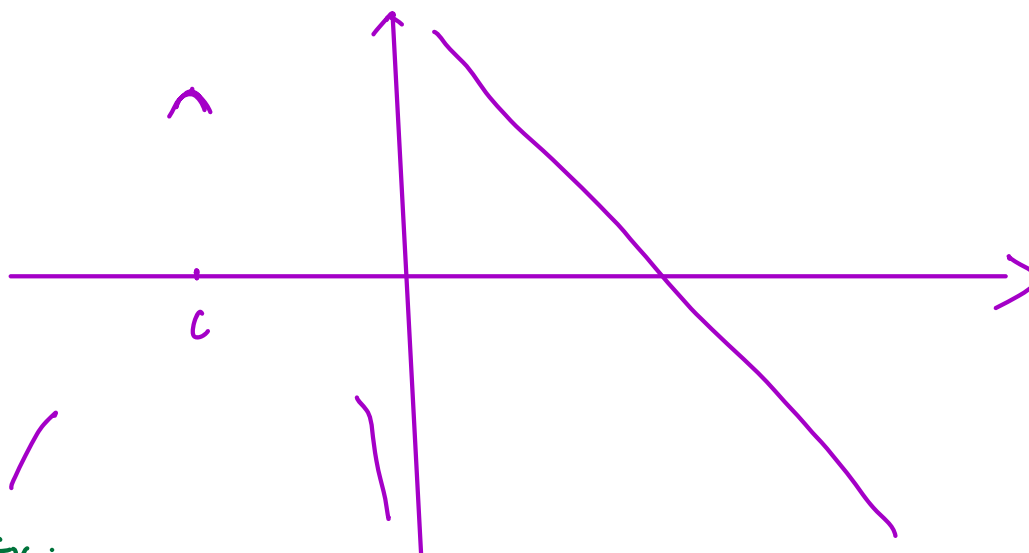
③ monotonicity: $f'(x) = -\frac{1}{x^2} - 2x$, $x \neq 0$.

Let $f'(c) = 0 \Rightarrow -\frac{1}{c^2} - 2c = -\frac{1}{c^2}(1 + 2c^3) = 0 \Rightarrow c = -(\frac{1}{2})^{\frac{1}{3}}$.

$-1 < c \Leftarrow -1 < -(\frac{1}{2})^{\frac{1}{3}} < -\frac{1}{2} \Leftarrow \frac{1}{2} < \frac{1}{2^{\frac{1}{3}}} < 1 \Leftarrow 1 < 2^{\frac{1}{3}} < 2$

$\therefore c = -(\frac{1}{2})^{\frac{1}{3}} < 0$

	$(-\infty, c)$	$(c, 0)$	$(0, \infty)$
f'	$f'(c-1) = -1 + 2 > 0$	$f'(-\frac{1}{2}) < 0$	$f'(1) < 0$
f	$\nearrow$	$\searrow$	$\searrow$



④ Concavity:

$$f''(x) = \frac{2}{x^3} - 2 = \frac{2}{x^3} (1 - x^3), \quad x \neq 0.$$

$$\text{Let } f''(c_2) = 0 \Rightarrow \frac{2}{c_2^3} (1 - (c_2)^3) = 0 \Rightarrow c_2 = 1.$$

	$(-\infty, \overset{-1}{\downarrow} 0)$	$(0, \overset{\frac{1}{2}}{\downarrow} 1)$	$(1, \overset{2}{\downarrow} \infty)$
f''	$f''(-1) < 0$	$f''(\frac{1}{2}) > 0$	$f''(2) < 0$
f	conc' down	conc' up	conc' down

Further:

(2) oblique: $\lim_{x \rightarrow \infty} \frac{f(x)}{x} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x} - x^2}{x} = \lim_{x \rightarrow \infty} \left(\frac{1}{x^2} - x \right) = -\infty.$

$\therefore$ There is no oblique asymptote as $x \rightarrow \infty$

($x \rightarrow -\infty$ is the same)

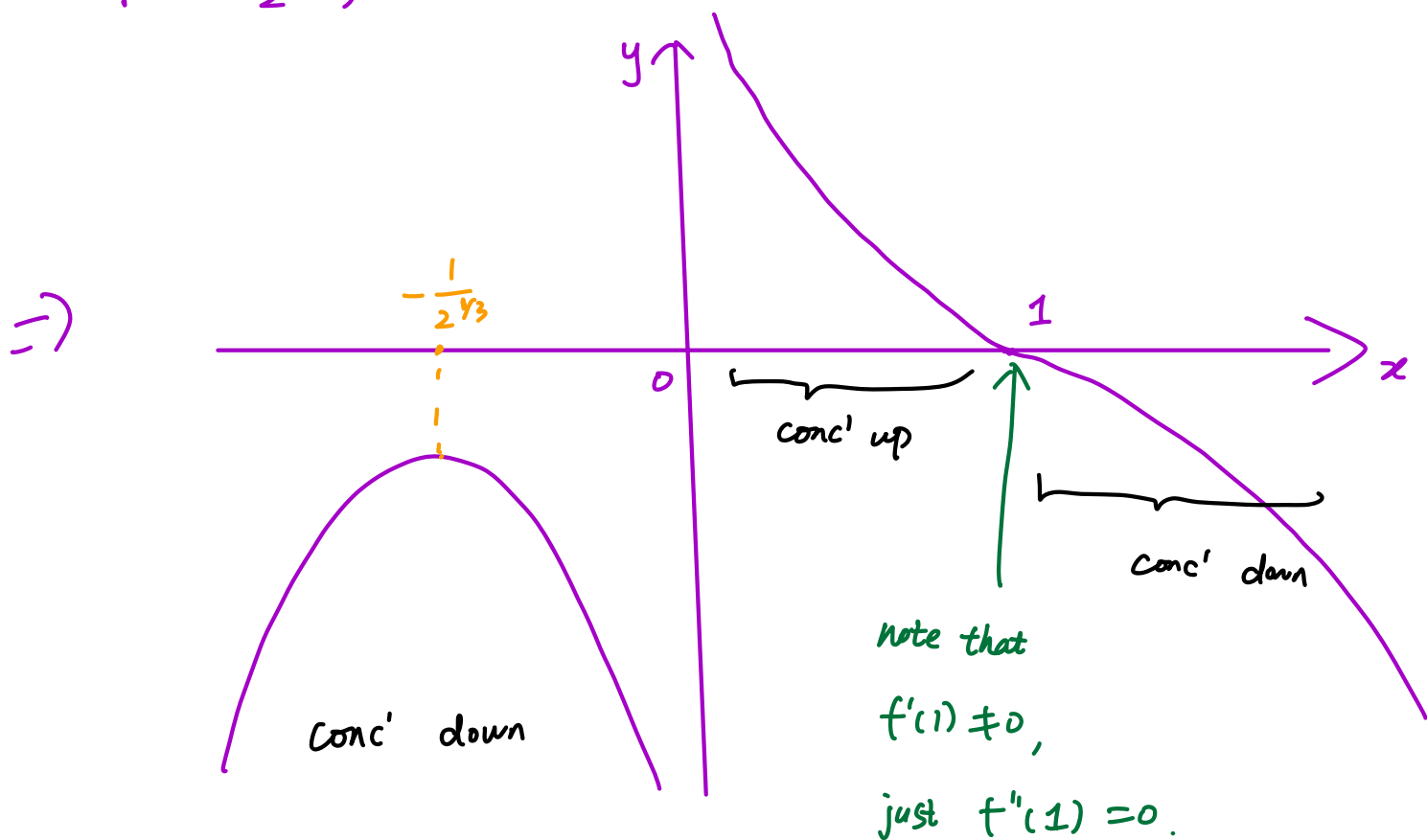
⑤ x - and y -intersects.

• Let $f(x)=0 \Rightarrow \frac{1}{x} - x^2 = 0 \Rightarrow x=1$.

• As 0 is not defined, there is no y -intersects

⑥ key points' value.

$$f\left(-\frac{1}{2^{1/3}}\right) = \dots = -2^{1/3} - 2^{-2/3} < 0.$$



See chap 4.4 for more examples.